

ME326 Group Project

TidyBot++ Collaborative Robotic Assistant

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Overview

1

Object Retrieval

"Find the apple and bring it here." Voice command triggers a full pipeline: scan the scene, locate the object, navigate to it, pick it up, and return.

2

Sequential Task

"Find the red apple and place it in the brown basket." Multi-step natural language commands that chain perception, navigation, and manipulation.

3

Drawer Articulation

Open a drawer in a cabinet using one arm, demonstrating dexterous manipulation of articulated objects.

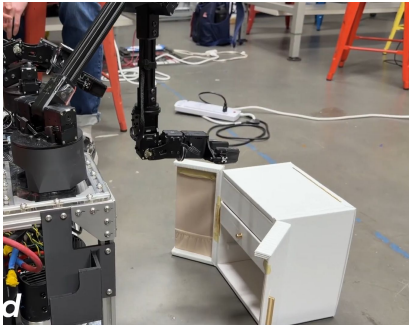
Task 3 Design Considerations

1

Handle Geometry

Are we using a knob? Pull lever?

This determines **grasp strategy**.

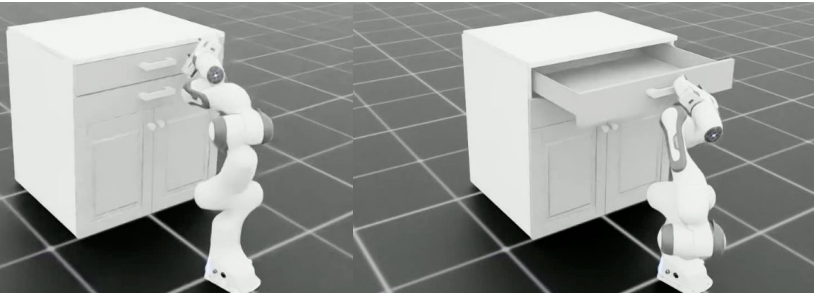


2

Required pull force

Static friction spike to start motion
Lower **kinetic friction** during sliding.

Is our robot controller able to accomplish the task?



3

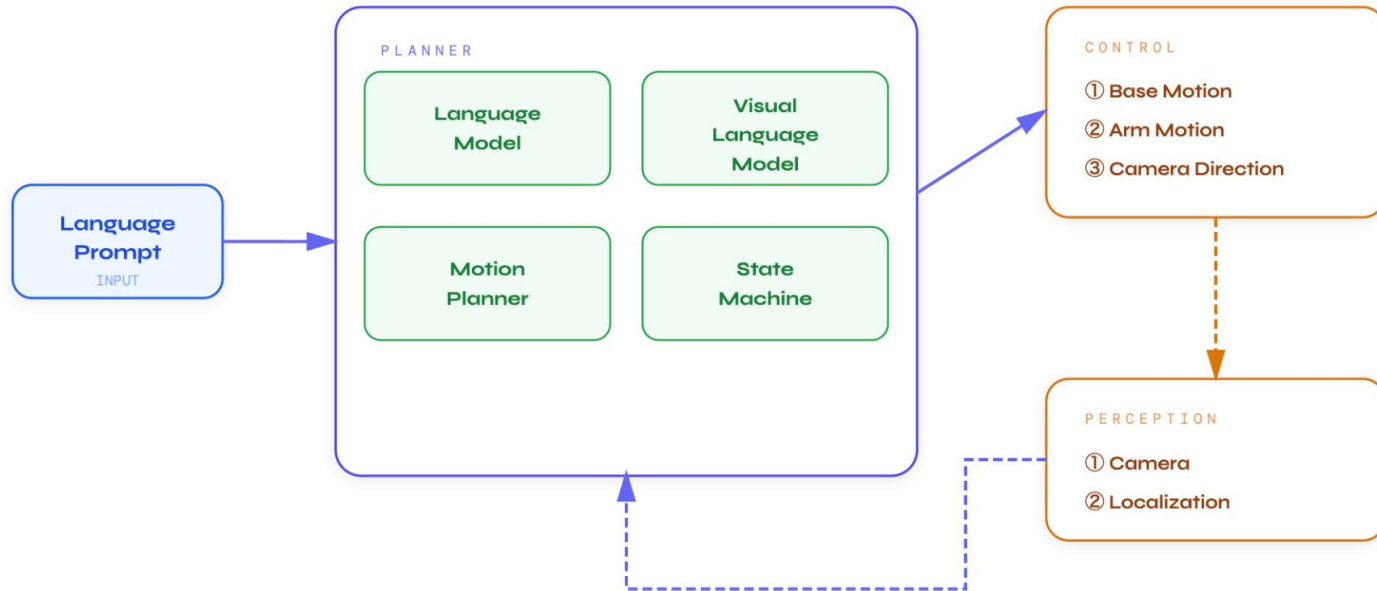
Direction of motion

Adds an **additional EE constraint**
Align pull vector with rail axis
Avoid lateral torque

May require:

- Visual localization
- Priors about task
- Tactile confirmation
- Force feedback

Method – High-Level Architecture

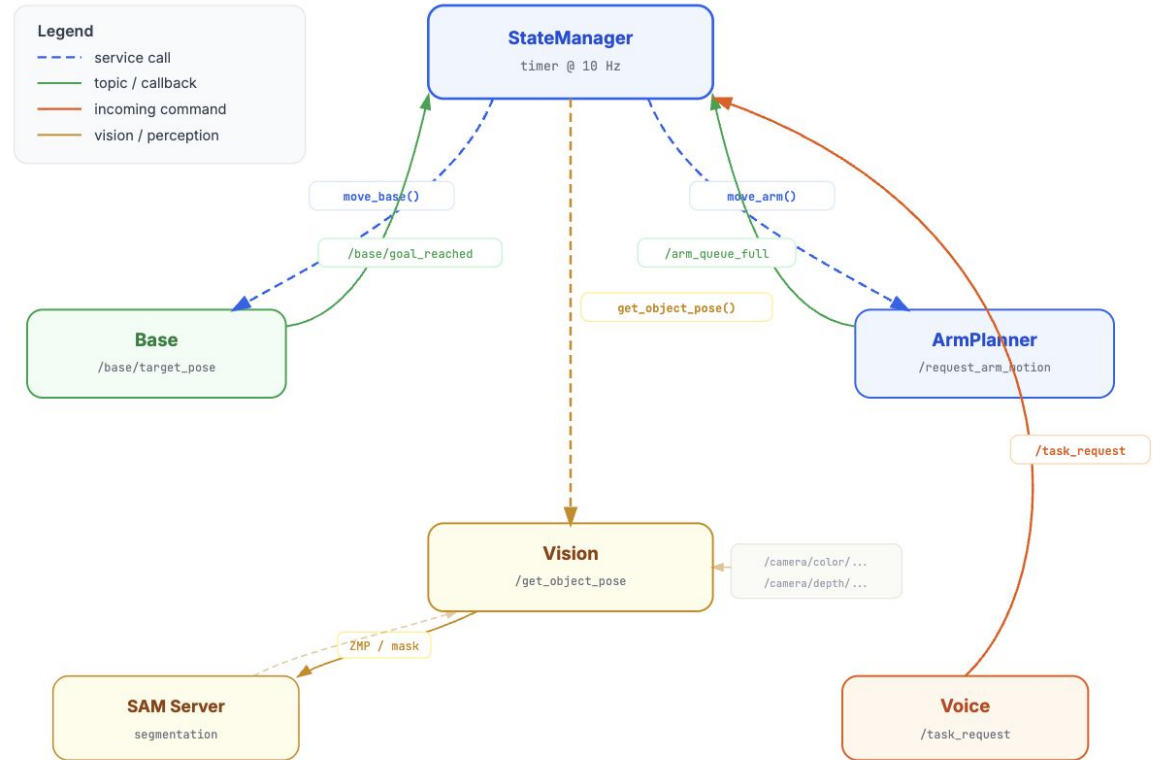


System Nodes & Services

Use of **services**, called by **State Manager**

Services are called **asynchronously**, and continually publish status

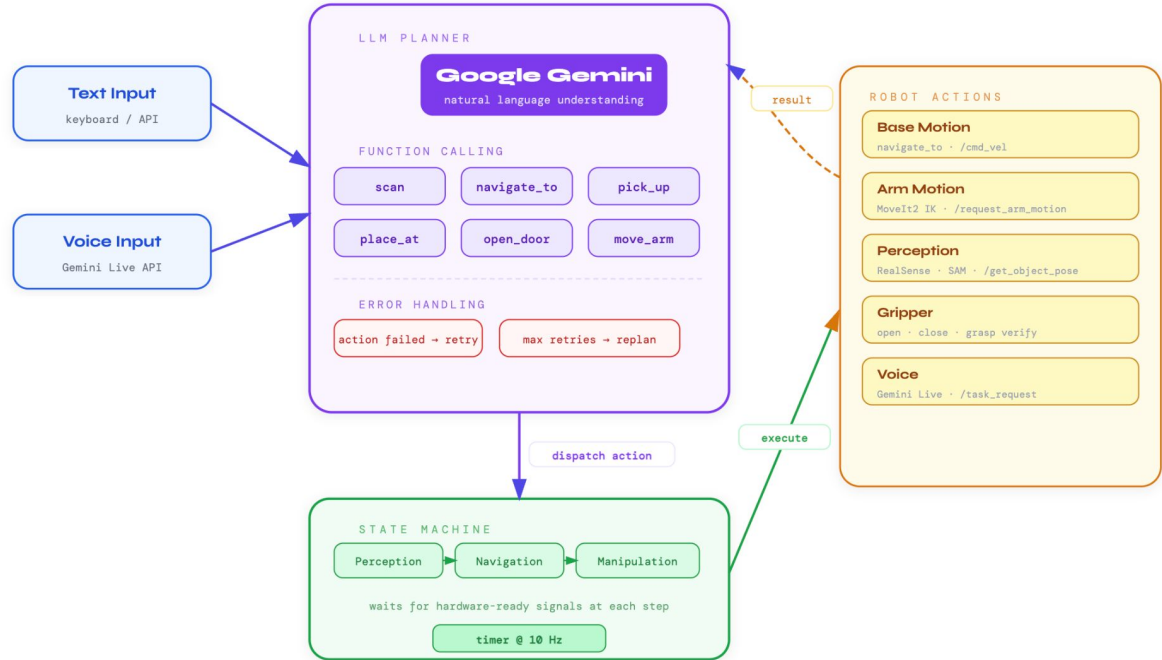
Service calls are better for intermittent messaging (vs publishers/subscribers)



Planning

-LLM Planner (Google Gemini
using tool calls)

-Voice & Text Input
-State Machine



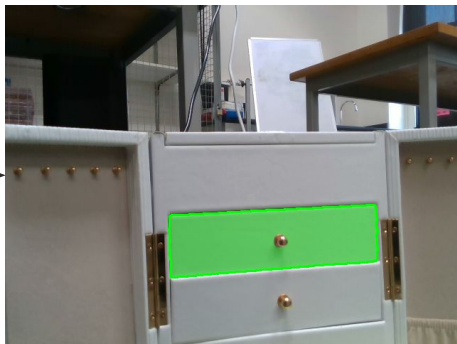
Perception Node

ROS2 service: given a prompt, returns a pose

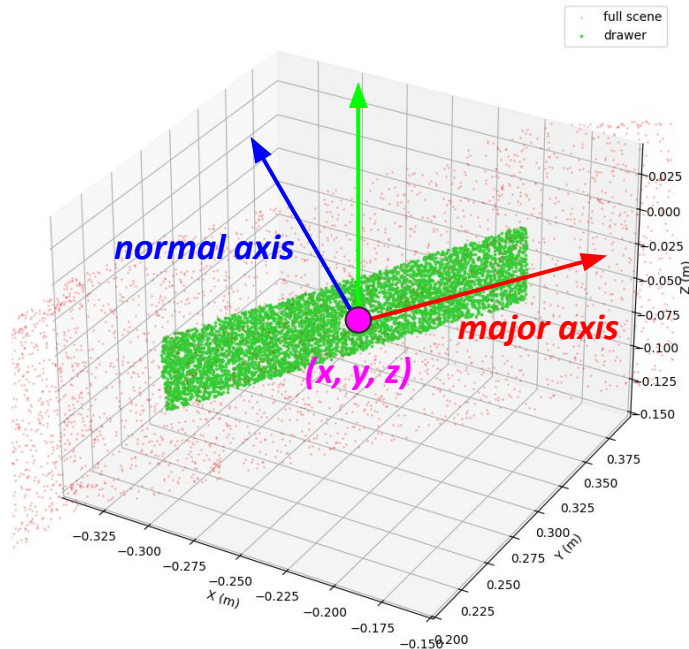


**Object detection +
masking via SAM3 server**

Prompt:
“drawer”



PCA analysis on segmented point cloud to obtain grasp pose



*Task 3 – additional “shiny gold knob”
prompt to obtain grasp position*

Arm Control

Built node to host a service with an action queue:

High-Level Controller

Grab

Release

Move

+ 6D Pose Message

Arm Control Queue

["reach", "close"]

["reach", "open"]

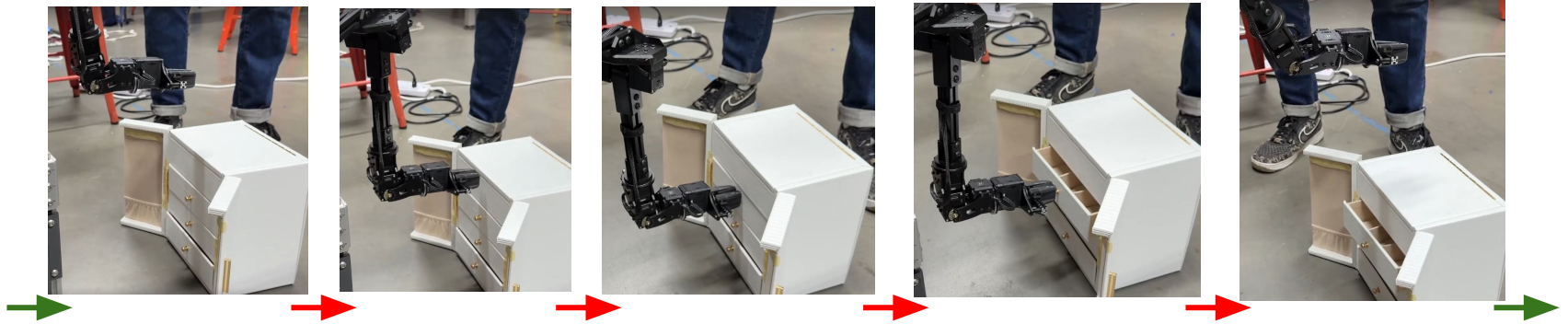
["reach"]

IK Solver

(and low level controller)

Arm Control

The action queue allows easier combination of waypoints:



Sent in five, non-blocking service calls from high-level planner!

Continually publish queue length, indicating when finished.

Base Control

Made a wrapper to host a service and add relative motion commands:

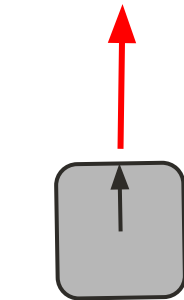
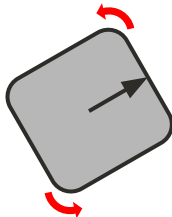
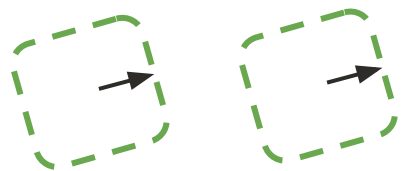
High-Level Controller Sends:

2D Target Pose
(x, y, theta)

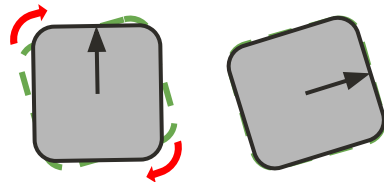
Relative/World (Bool)



Goal Pose:



TidyBot Base

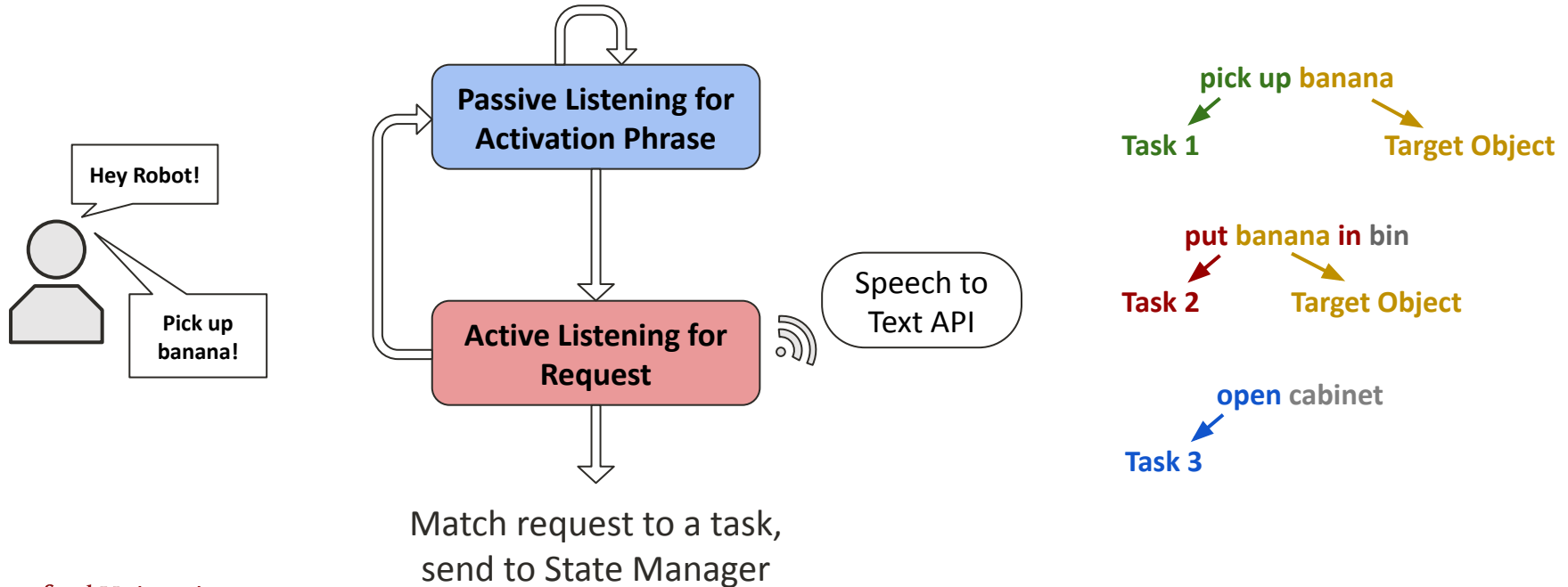


Phoenix Base Node
performs 3-phase
position control,
given a goal pose

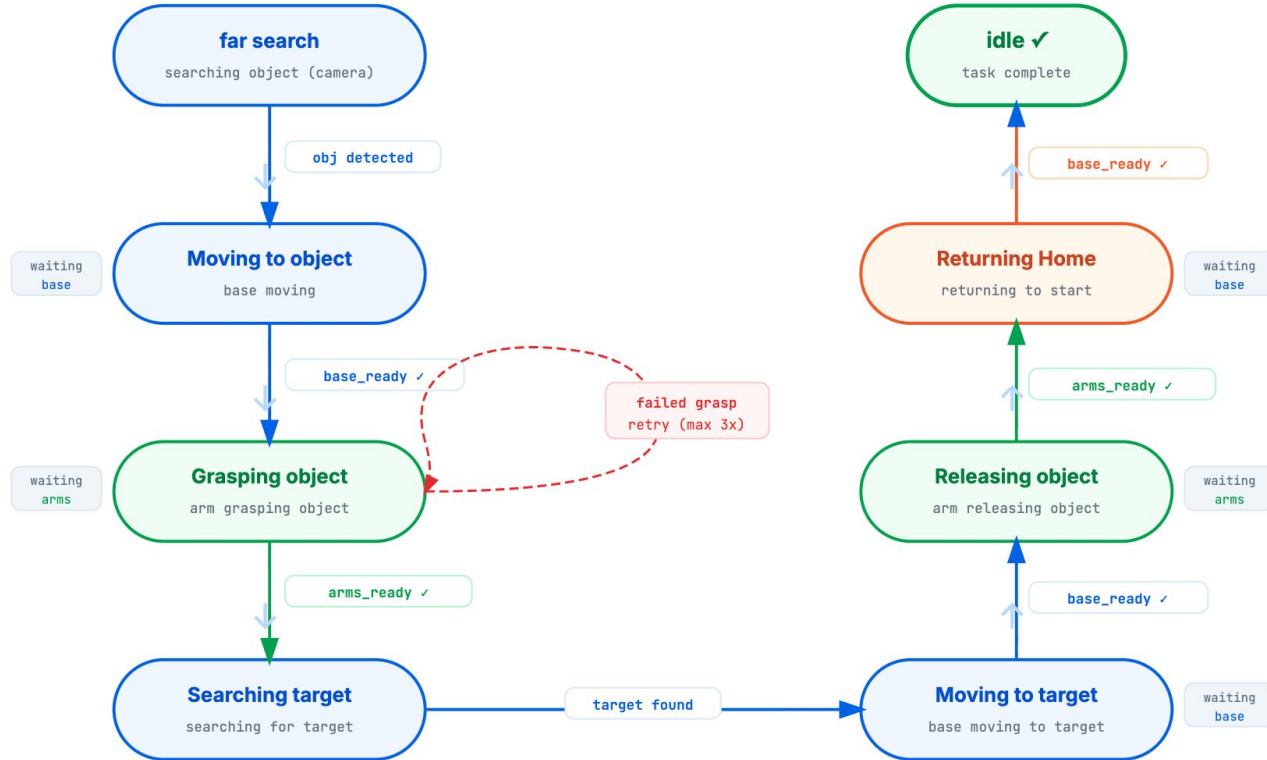
Publishes when goal
is reached

Voice Control

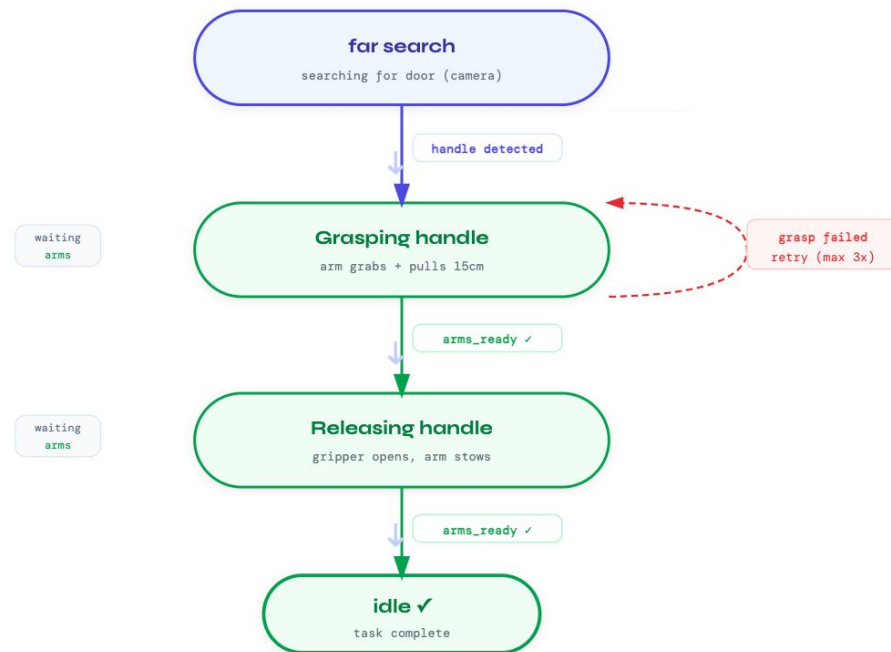
Made a ROS2 Node with Google's Cloud Speech-to-Text API:



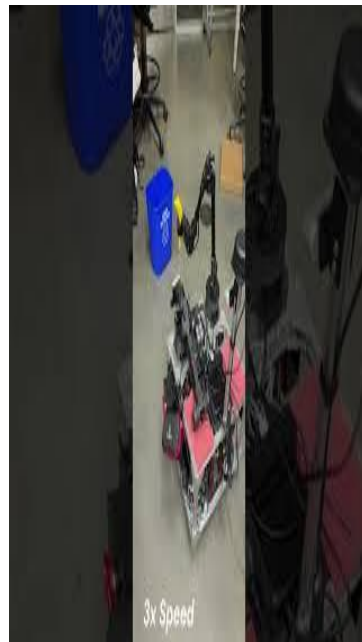
Pick & Place State Machine



Drawer Articulation State Machine



Results- Tasks 1 & 2



Results- Task 3



Future work

Robust Grasping

Integrate GraspAnything for 6-DOF grasp pose prediction.

Force feedback, tactile sensors.

Bimanual Complexity

Extend dual-arm coordination to folding, packing, and tool use.

Dynamic Replanning

Real-time failure detection via **memory** or VLA planner to auto-recover if failed

Human-Robot Collab

Hand objects to humans, respond to gestures, operate in shared spaces.

Intent estimation.

SLAM

Safety-critical and robust SLAM with onboard LiDAR for obstacle avoidance and planning

Learning from Demo

Learn manipulation skills from human demonstrations

Behavior cloning, **imitation learning**, ACT, Diffusion policy.



Questions?

Thank you